Sri Lanka Institute of Information Technology

IT3010 – Network Design & Management (NDM)

Practical 02 - Report

WE.3Y.S2.2.1

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Introduction

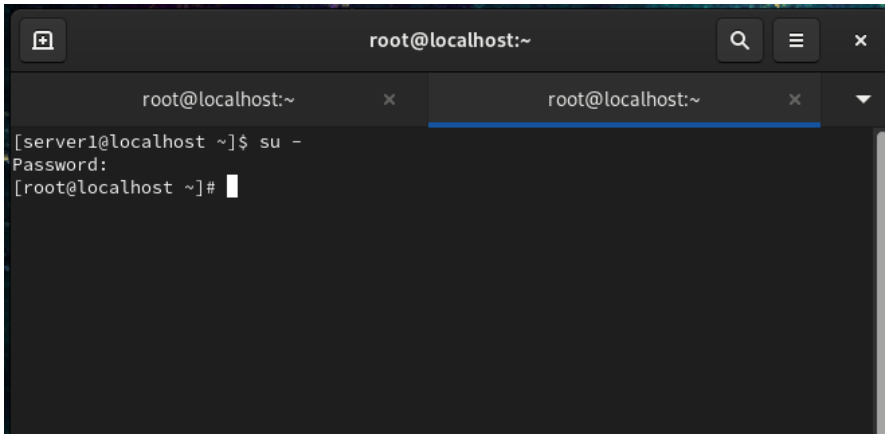
The Dynamic Host Configuration Protocol (DHCP) is a critical network management protocol used on UDP/IP networks. It allows a DHCP server to dynamically assign IP addresses and other essential network configuration parameters to each device on a network. This automation ensures that devices can communicate efficiently with other IP networks without the need for manual configuration by a network administrator.

In this laboratory experiment, the primary objective is to install and configure a DHCP server within a CentOS environment. To verify the successful deployment of the service, a Fedora client will be used to request and receive network configurations from the configured CentOS server.

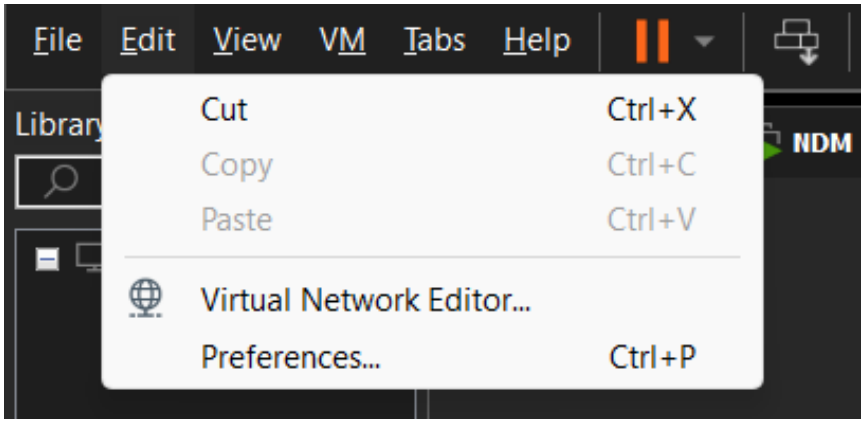
Installing and configuring DHCP

Disable the DHCP settings in VMnet 2.

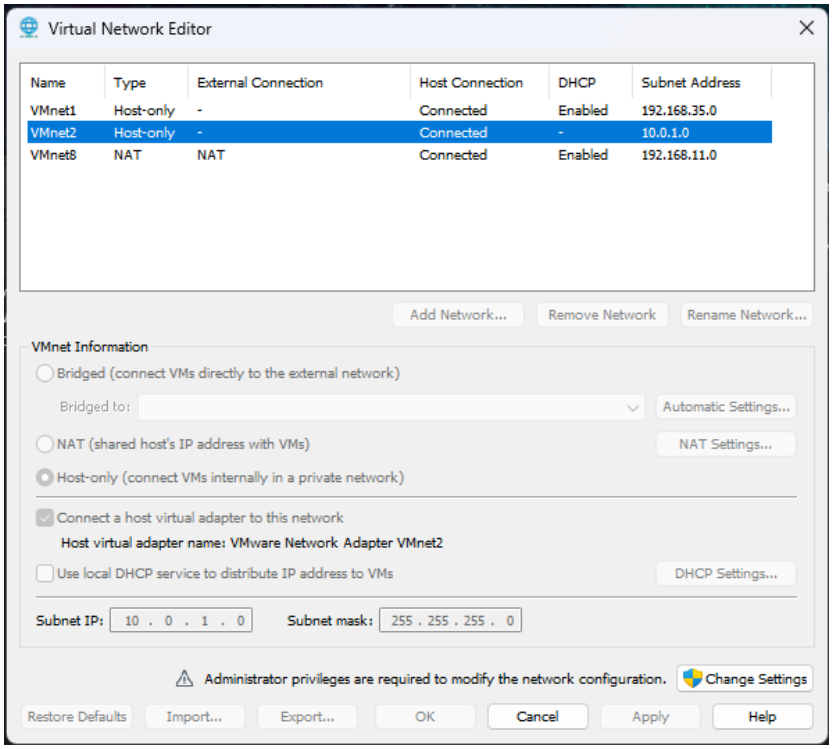
Step 1



Step 2

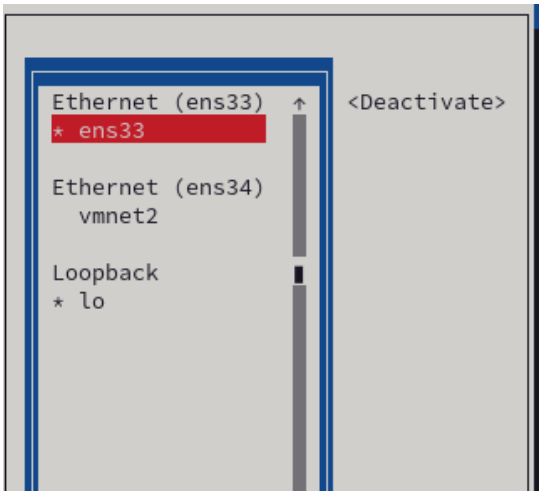


Step 3



To install DHCP server on CentOS

Step 1



Activate NAT connection and disable vmnet2

Step 2

```
root@localhost:~
[root@localhost ~]# dnf install -y dhcp-server
CentOS Stream 9 - BaseOS                                        283 kB/s | 8.9 MB     00:32
CentOS Stream 9 - AppStream                                     848 kB/s |  27 MB     00:32
CentOS Stream 9 - Extras packages                               4.8 kB/s |  20 kB     00:04
Last metadata expiration check: 0:00:01 ago on Wed 18 Feb 2026 08:16:49 PM +0530.
Dependencies resolved.
================================================================================================
 Package          Architecture     Version                  Repository              Size
================================================================================================
Installing:
 dhcp-server      x86_64           12:4.4.2-20.b1.el9       baseos                 1.2 M
Installing dependencies:
 dhcp-common      noarch           12:4.4.2-20.b1.el9       baseos                 127 k

Transaction Summary
================================================================================================
Install  2 Packages

Total download size: 1.3 M
Installed size: 4.2 M
Downloading Packages:
(1/2): dhcp-common-4.4.2-20.b1.el9.noarch.rpm                   70 kB/s | 127 kB     00:01
(2/2): dhcp-server-4.4.2-20.b1.el9.x86_64.rpm                  355 kB/s | 1.2 MB     00:03
------------------------------------------------------------------------------------------------
Total                                                          354 kB/s | 1.3 MB     00:03
CentOS Stream 9 - BaseOS                                       1.0 MB/s | 1.6 kB     00:00
Importing GPG key 0x8483C65D:
 Userid     : "CentOS (CentOS Official Signing Key) <security@centos.org>"
 Fingerprint: 99DB 70FA E1D7 CE22 7FB6 4882 05B5 55B3 8483 C65D
 From       : /etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
```

This command installs the DHCP server package on the CentOS system.

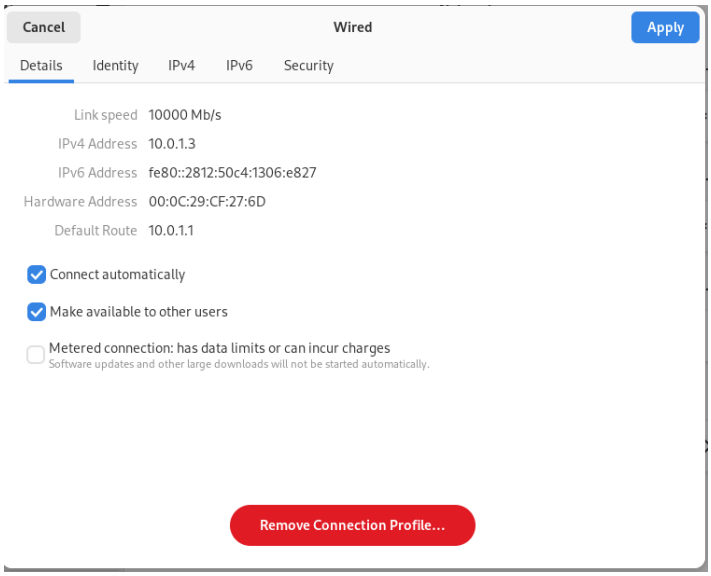
Step 3

```
Complete!
[root@localhost ~]# ping www.google.com
PING www.google.com (142.251.157.119) 56(84) bytes of data.
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=1 ttl=128 time=66.7 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=2 ttl=128 time=81.6 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=3 ttl=128 time=81.7 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=4 ttl=128 time=81.3 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=5 ttl=128 time=69.1 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=6 ttl=128 time=74.7 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=7 ttl=128 time=74.3 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=8 ttl=128 time=68.6 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=9 ttl=128 time=70.6 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=10 ttl=128 time=67.0 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=11 ttl=128 time=64.3 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=12 ttl=128 time=69.8 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=13 ttl=128 time=83.2 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=14 ttl=128 time=78.2 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=15 ttl=128 time=76.6 ms
64 bytes from 142.251.157.119 (142.251.157.119): icmp_seq=16 ttl=128 time=81.4 ms
```

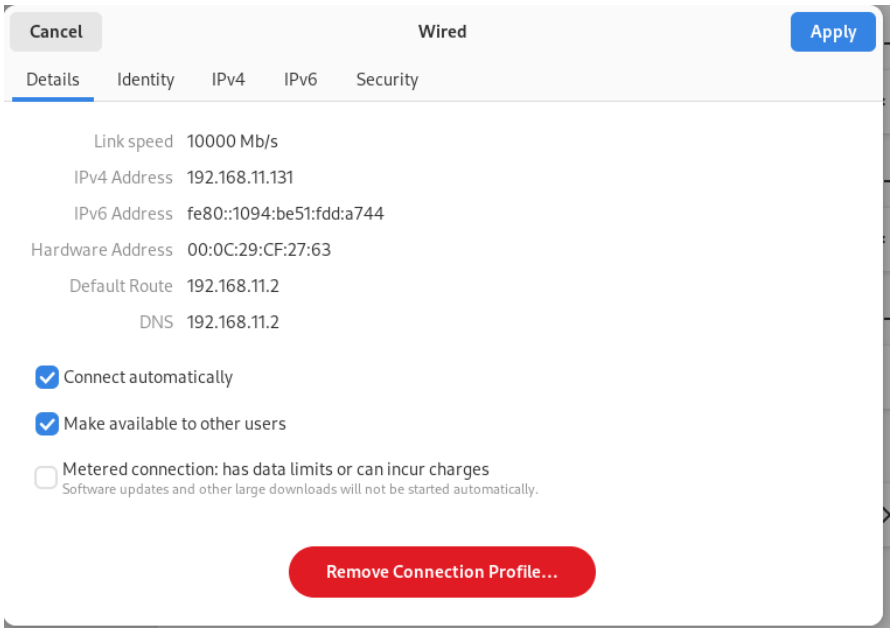
This command checks whether the system is connected to the interne

Client - Configuring DHCP settings. Now we need to mention the interface details, which is going to be the DHCP interface.

Step 1



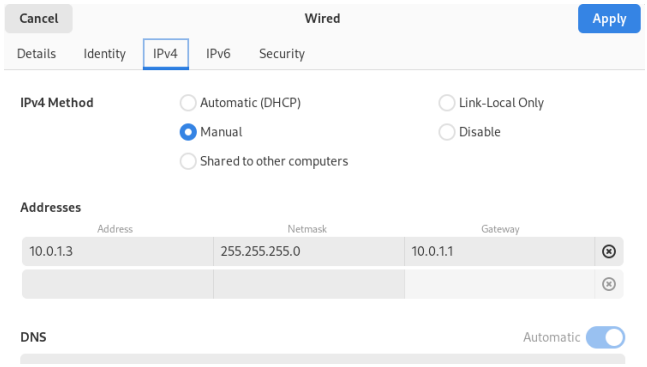
This is manual assign ip address – vmnet2 (ens192)



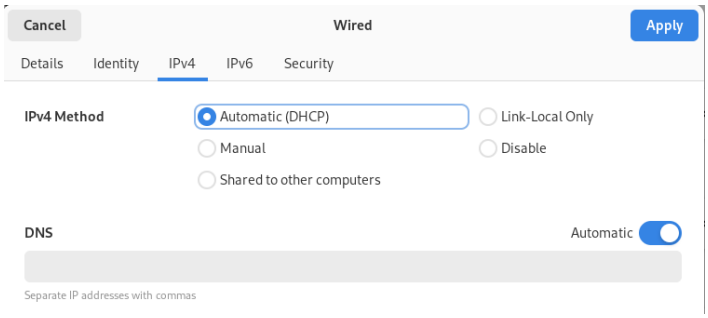
This is NAT address (ens160)

Disable in manual Ip address – vmnet2

Step 1

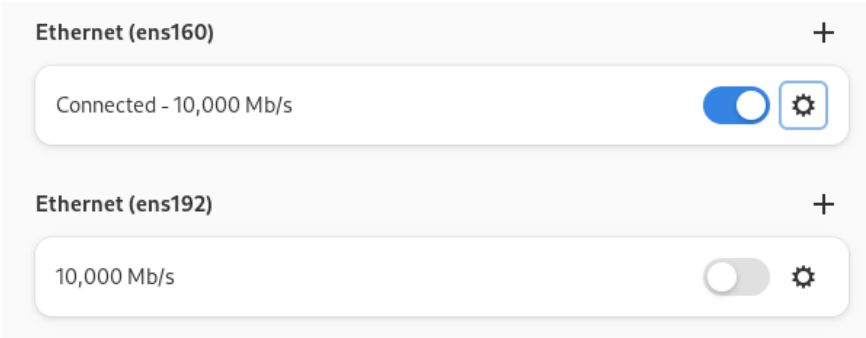


Step 2



Remember after doing previous steps, refresh network.

Step 3



Disable in NAT.

Server - Configuring DHCP settings.

Step 1

```
[server1@localhost ~]$ su -
Password:
[root@localhost ~]# vi /etc/sysconfig/dhcpd
```

This command opens the DHCP configuration file to set the network interface for the DHCP server.

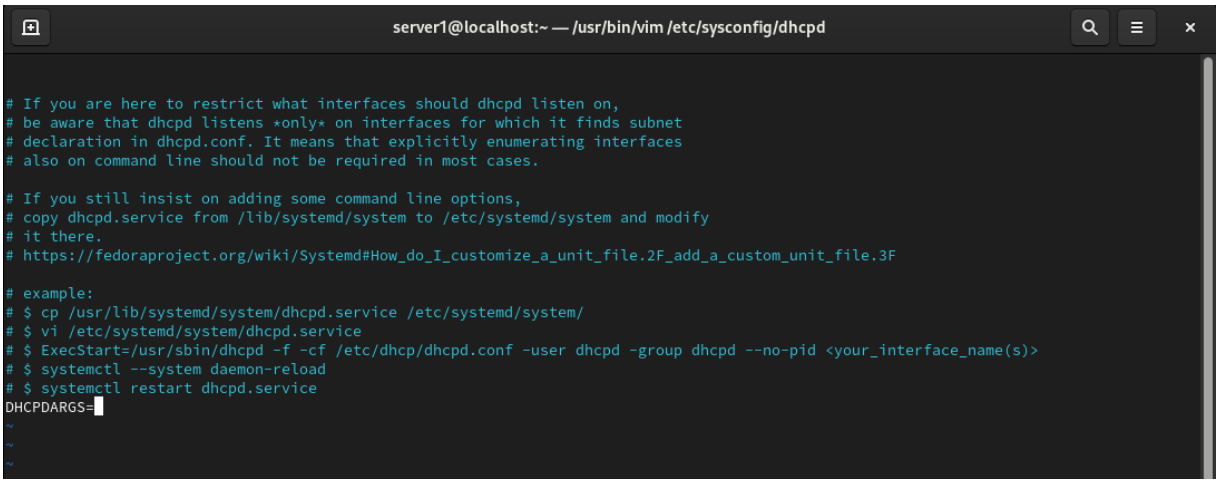
Step 2

```
server1@localhost:~ — /usr/bin/vim /etc/sysconfig/dhcpd

# If you are here to restrict what interfaces should dhcpd listen on,
# be aware that dhcpd listens *only* on interfaces for which it finds subnet
# declaration in dhcpd.conf. It means that explicitly enumerating interfaces
# also on command line should not be required in most cases.

# If you still insist on adding some command line options,
# copy dhcpd.service from /lib/systemd/system to /etc/systemd/system and modify
# it there.
# https://fedoraproject.org/wiki/Systemd#How_do_I_customize_a_unit_file.2F_add_a_custom_unit_file.3F

# example:
# $ cp /usr/lib/systemd/system/dhcpd.service /etc/systemd/system/
# $ vi /etc/systemd/system/dhcpd.service
# $ ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid <your_interface_name(s)>
# $ systemctl --system daemon-reload
# $ systemctl restart dhcpd.service
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --                                                    1,1           All
```

This file is edited to specify which network interface the DHCP server should use.

When you press I you go to insert mode from there you can draw commands like bellow

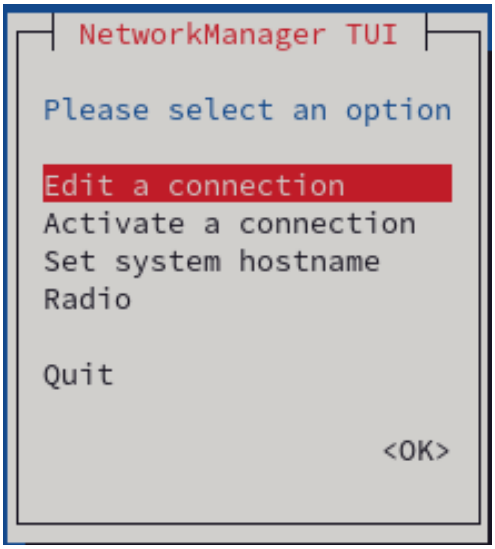
You can use the line DHCPDARGS to do that



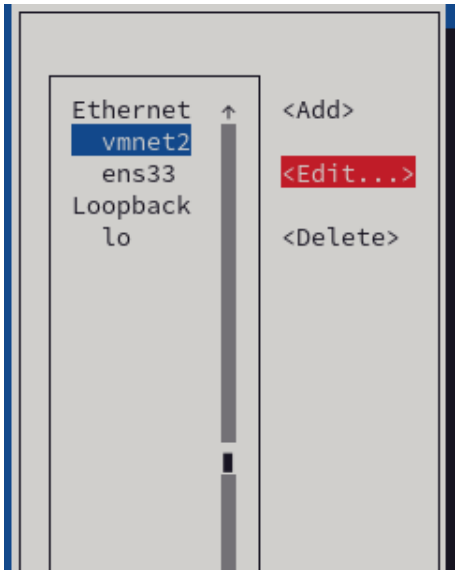
Here we need to find our network Interface for the DHCPDARGS=…….. Command,

The steps for that are below,

Step 1



Step 2



Step 3

```
                          Edit Connection

        Profile name vmnet2
              Device ens34 (00:0C:29:1C:71:AD)

= ETHERNET                                                  <Show>
= 802.1X SECURITY                                           <Show>

= IPv4 CONFIGURATION <Manual>                               <Hide>
          Addresses 10.0.1.2/24          <Remove>
                    <Add...>
            Gateway 10.0.1.1
        DNS servers <Add...>
     Search domains <Add...>

            Routing (No custom routes) <Edit...>
  [ ] Never use this network for default route
  [ ] Ignore automatically obtained routes
  [ ] Ignore automatically obtained DNS parameters

  [ ] Require IPv4 addressing for this connection

= IPv6 CONFIGURATION <Automatic>                            <Show>

[X] Automatically connect
[X] Available to all users

                                                   <Cancel> <OK>
```

Here we can see the profile name vmnet2 and the device name ens34, This is the device name we are giving here

```
# $ systemctl restart dhcpd.service
DHCPDARGS=ens34
~
```

Step 3

```
server1@localhost:~ — /usr/bin/vim /etc/sysconfig/dhcpd

# If you are here to restrict what interfaces should dhcpd listen on,
# be aware that dhcpd listens *only* on interfaces for which it finds subnet
# declaration in dhcpd.conf. It means that explicitly enumerating interfaces
# also on command line should not be required in most cases.

# If you still insist on adding some command line options,
# copy dhcpd.service from /lib/systemd/system to /etc/systemd/system and modify
# it there.
# https://fedoraproject.org/wiki/Systemd#How_do_I_customize_a_unit_file.2F_add_a_custom_unit_file.3F

# example:
# $ cp /usr/lib/systemd/system/dhcpd.service /etc/systemd/system/
# $ vi /etc/systemd/system/dhcpd.service
# $ ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid <your_interface_name(s)>
# $ systemctl --system daemon-reload
# $ systemctl restart dhcpd.service
DHCPDARGS=ens34
~
~
~
```

Save steps = Exit insert mode (Using Esc key) and type **: wq!** Command, press enter.

Copy the sample dhcp configuration file to /etc/dhcp/ directory

Step 1

```
root@localhost:~

[server1@localhost ~]$ su -
Password:
[root@localhost ~]# vi /etc/sysconfig/dhcpd
[root@localhost ~]# cp /usr/share/doc/dhcp-server/dhcpd.conf.example /etc/dhcp/dhcpd.conf
```

The system switches to the root user and copies the sample DHCP configuration file to the main configuration folder.

Step 2

```
root@localhost:~
[server1@localhost ~]$ su -
Password:
[root@localhost ~]# vi /etc/sysconfig/dhcpd
[root@localhost ~]# cp /usr/share/doc/dhcp-server/dhcpd.conf.example /etc/dhcp/dhcpd.conf
cp: overwrite '/etc/dhcp/dhcpd.conf'? yes
[root@localhost ~]#
```

Step 3

```
root@localhost:~
[server1@localhost ~]$ su -
Password:
[root@localhost ~]# vi /etc/sysconfig/dhcpd
[root@localhost ~]# cp /usr/share/doc/dhcp-server/dhcpd.conf.example /etc/dhcp/dhcpd.conf
cp: overwrite '/etc/dhcp/dhcpd.conf'? yes
[root@localhost ~]# vi /etc/dhcp/dhcpd.conf
```

The sample DHCP file is copied and opened to configure IP address settings.

Step 4

```
root@localhost:~
# dhcpd.conf

# Sample configuration file for ISC dhcpd
#

# option definitions common to all supported networks...
option domain-name "example.org";
option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 600;
max-lease-time 7200;

# Use this to enble / disable dynamic dns updates globally.
#ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
#authoritative;

# Use this to send dhcp log messages to a different log file (you also
# have to hack syslog.conf to complete the redirection).
log-facility local7;

# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.

subnet 10.152.187.0 netmask 255.255.255.0 {
}

# This is a very basic subnet declaration.

subnet 10.254.239.0 netmask 255.255.255.224 {
  range 10.254.239.10 10.254.239.20;
  option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
}

# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.

subnet 10.254.239.32 netmask 255.255.255.224 {
  range dynamic-bootp 10.254.239.40 10.254.239.60;
  option broadcast-address 10.254.239.31;
"/etc/dhcp/dhcpd.conf" 104L, 3266B
```

This is DHCP editor.

If this DHCP server is the official DHCP server for the local network, you should uncomment the following line:

Step 1

```
# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
#authoritative;
```

Step 2

```
# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;
```

Define the sunbet, range of ip addresses, domain and domain name servers

Step 1

```
# A slightly different configuration for an internal subnet.
subnet 10.5.5.0 netmask 255.255.255.224 {
  range 10.5.5.26 10.5.5.30;
  option domain-name-servers ns1.internal.example.org;
  option domain-name "internal.example.org";
  option routers 10.5.5.1;
  option broadcast-address 10.5.5.31;
  default-lease-time 600;
  max-lease-time 7200;
}
```

Step 2

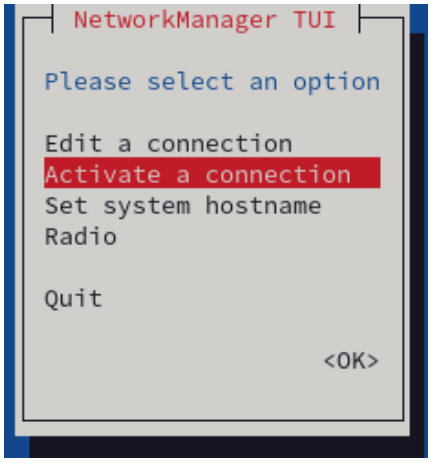
```
# A slightly different configuration for an internal subnet.
subnet 10.0.1.0 netmask 255.255.255.0 {
  range 10.0.1.25 10.0.1.50;
  option domain-name-servers ns1.internal.example.org;
  option domain-name "internal.example.org";
  option routers 10.0.1.1;
  option broadcast-address 10.0.1.255;
  default-lease-time 600;
  max-lease-time 7200;
}
```

- After making all the changes you want, save and close the file. Be mindful that if you have another unused entries on the dhcpd.conf file, comment them. Otherwise, you'll have issues while starting dhcpd service.

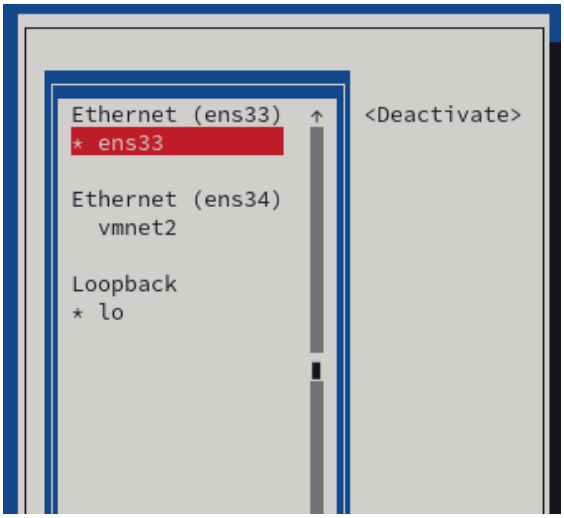
The dhcpd service and make it to start automatically on every reboot.

1st one , now go to Active connections turn off ens33 and turn on vmnet2,The steps are below.

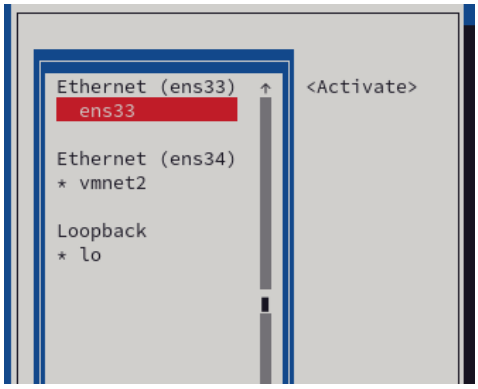
Step 1



Step 2



Step 3

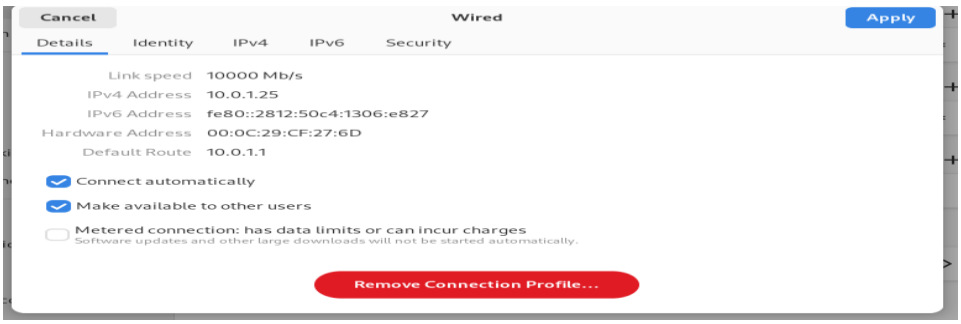


Start DHCP

```
[root@localhost ~]# service dhcpd start
Redirecting to /bin/systemctl start dhcpd.service
[root@localhost ~]#
```

If you want to start up the DHCP server at logon to the server session use

```
[root@localhost ~]# systemctl status dhcpd
● dhcpd.service - DHCPv4 Server Daemon
     Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-02-18 23:03:57 +0530; 2min 42s ago
       Docs: man:dhcpd(8)
             man:dhcpd.conf(5)
   Main PID: 38793 (dhcpd)
     Status: "Dispatching packets..."
      Tasks: 1 (limit: 10291)
     Memory: 6.0M (peak: 6.3M)
        CPU: 47ms
     CGroup: /system.slice/dhcpd.service
             └─38793 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid ens34

Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Source compiled to use binary-leases
Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Wrote 0 class decls to leases file.
Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Wrote 0 deleted host decls to leases file.
Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Wrote 0 new dynamic host decls to leases file.
Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Wrote 0 leases to leases file.
Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Listening on LPF/ens34/00:0c:29:1c:71:ad/10.0.1.0/24
Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Sending on   LPF/ens34/00:0c:29:1c:71:ad/10.0.1.0/24
Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Sending on   Socket/fallback/fallback-net
Feb 18 23:03:57 localhost.localdomain dhcpd[38793]: Server starting service.
Feb 18 23:03:57 localhost.localdomain systemd[1]: Started DHCPv4 Server Daemon.
[root@localhost ~]# S
```

Checking the status of DHCP server and client.- ifconfig

```
ens192: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.1.25  netmask 255.255.255.0  broadcast 10.0.1.255
        inet6 fe80::2812:50c4:1306:e827  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:cf:27:6d  txqueuelen 1000  (Ethernet)
        RX packets 437  bytes 63862 (62.3 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2912  bytes 760168 (742.3 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```